

Client-Server Architecture

Clients: Devices or software (like browsers or apps) that request services or data.

Servers: Powerful computers or software that process requests and send back data.

Benefits: Centralized control, easier updates, better security, and efficient resource sharing.

Use Cases: Websites, online banking, email systems, multiplayer games.

Critical Components: Clients and Servers.

Types of Networks

Intranet: A private network inside an organization for employees only.

Extranet: An extension of the intranet that allows limited access to outsiders (partners, customers).

Internet: The global network connecting millions of computers worldwide.

Types of Websites

Static Websites: Fixed content, doesn't change unless updated manually (e.g., company info pages).

Dynamic Websites: Content changes automatically based on user input or databases (e.g., e-commerce sites).

Interactive Websites: Allow users to actively engage (e.g., online games, Google Docs).

Local Web Server Setup

Manual Installation: Install each component separately (Apache, PHP, MySQL).

XAMPP: Cross-platform package (Windows, Linux, Mac) with Apache, MySQL, PHP, Perl.

WAMP: Windows-based package (Windows, Apache, MySQL, PHP).

MAMP: Mac-based package (Mac, Apache, MySQL, PHP).

LAMP: Linux-based stack (Linux, Apache, MySQL, PHP).

Purpose: To test and develop websites locally before putting them online.

Key Q&A

1. **Name Two Critical Components of Client-Server Architecture.**
Clients & Servers.
2. **Explain The Role of Servers in Client-Server Architecture.**
Handle requests, process data, send results.
3. **Define an Intranet.**
Private network for internal use.
4. **What is The Purpose of an Extranet?**
Share limited access with outsiders.
5. **Explain the concept of an extranet.**
Intranet + controlled external access.
6. **What is the significance of the TCP/IP protocol suite in network communication?**
Standard rules for reliable communication on the Internet.
7. **What are some examples of interactive websites?**
Online games, Google Docs, educational apps.
8. **What is the purpose of a local web server?**
Test websites on your computer before going live.
9. **Name two popular software packages for setting up a local web server environment.**
XAMPP, WAMP.
10. **Which components are typically included in a LAMP stack?**
Linux, Apache, MySQL, PHP.

What is a Web Browser?

A **web browser** is a software/tool/program that let us **access and view websites on the internet**.

Examples: Chrome, Firefox, Edge.

Types of Web Browsers (Based on Focus)

(Instead of overcomplicating it, just remember these categories:)

1. Standard Browsers

- Used for everyday browsing
- Fast, user-friendly

Examples: Chrome, Edge, Firefox, Safari

2. Privacy-Focused Browsers

- Protect your data
- Block trackers and ads

Examples: Brave, Tor Browser

3. Lightweight / Specialized Browsers

- Use fewer resources
- Offer unique tools or customization

Example: Opera

Popular Browsers & Their Features

Now let's break down each one in a way that we can explain confidently.

Mozilla Firefox

Definition:

A free, open-source browser focused on privacy and customization.

Key Features:

- Strong privacy protection
- Lots of extensions (add-ons)
- Highly customizable
- Not owned by a big tech ad company (important point)

Think: "Privacy + Freedom"



Google Chrome

Definition:

The most popular web browser developed by Google.

Key Features:

- Very fast
- Simple and clean interface
- Syncs with your Google account
- Huge extension library

Think: "Speed + Popularity"



Safari

Definition:

Apple's browser, mainly used on iPhone, iPad, and Mac.

Key Features:

- Optimized for Apple devices
- Energy efficient (saves battery)
- Smooth performance
- Strong privacy controls

Think: "Best for Apple users"



Opera

Definition:

A feature-rich browser with built-in tools.

Key Features:

- Built-in VPN
- Ad blocker
- Sidebar apps (WhatsApp, Messenger)
- Lightweight

Think: “All-in-one browser”



Microsoft Edge

Definition:

Microsoft’s modern browser (replaced Internet Explorer).

Key Features:

- Fast and efficient
- Works well with Windows
- Built-in PDF reader
- Good security

Think: “Best for Windows users”



Brave

Definition:

A browser focused heavily on privacy and blocking ads.

Key Features:

- Blocks ads automatically
- Blocks trackers
- Faster loading (because no ads)
- Optional crypto rewards system

Think: “Privacy + Speed”



Tor Browser

Definition:

A browser designed for **anonymous browsing**.

Key Features:

- Hides your identity
- Encrypts your traffic
- Access to deep web (.onion sites)
- Very strong privacy

Think: “Maximum anonymity (but slower)”

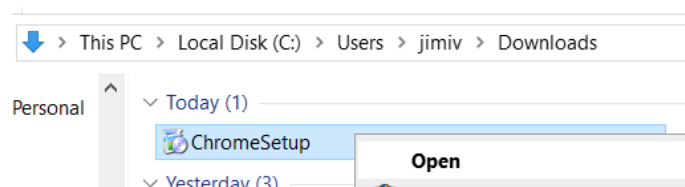
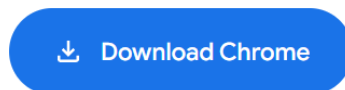
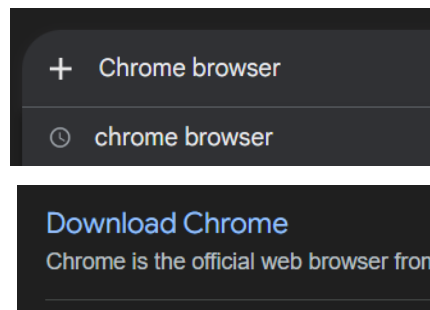


How to install Chrome in your Computer?

1. Open the default browser
2. Search “chrome browser”
3. Click on “download chrome browser”
4. Locate the downloaded file and install

Shorter Version:

Open browser → Search “Chrome” → Download → Install



Cross-Browser Compatibility

Cross-browser compatibility means a website works properly on **all browsers** (Chrome, Firefox, Edge, Safari, etc.)

Same design. Same features. Same experience.

Let's say we build a website once, but people open it in different browsers.

If it looks broken in one browser = **Bad**

If it works everywhere = **Cross-Browser Compatible**

Why It Matters

Not everyone uses the same browser.

Some use Chrome

Some use Firefox

Some still Edge

The website must behave consistently for all users.

True/False Answers (Simple & Clear)

1. Web browsers interpret HTML, CSS, and JavaScript to display webpages.

True

Why: Browsers read these languages and turn them into what we see on screen.

2. Opera offers a built-in ad blocker and free VPN.

True

Why: Opera comes with both features already included, no extra setup needed.

3. Cross-browser compatibility refers to the ability of a website to function consistently across different web browsers.

True

Why: A good website should look and work the same on Chrome, Firefox, Edge, etc.

4. CSS resets or normalization techniques eliminate inconsistencies in default browser styles.

True

Why: Different browsers style things differently, this makes them behave the same.

5. The Inspect Element tool allows live editing of the Document Object Model (DOM).

True

Why: You can change text, styles, or layout instantly (only on your screen).

6. The Inspect Element tool provides a network panel to monitor network requests made by a web page.

True

Why: You can see what files (images, scripts, data) are being loaded.

7. The Inspect Element tool is primarily used by website visitors to view the source code of a webpage.

False

Why: It's mainly for **developers**, not regular users, even though anyone *can* use it.

One-Line Summary (Bullet Version)

Browsers **render**, developers **debug**, and compatibility ensures everything **works everywhere**.

Short QnA:

- 1. What is a web browser?**
→ A program used to access and view websites.
- 2. Which web browser is developed by the Mozilla Foundation?**
→ Firefox.
- 3. Which web browser is the default browser for Apple devices?**
→ Safari.
- 4. Which web browser is based on the Chromium project and is the default browser for Windows 10?**
→ Microsoft Edge.
- 5. What is the purpose of feature detection in achieving cross-browser compatibility?**
→ To check if a browser supports a feature before using it.
- 6. What can developers do with the console in the Inspect Element tool?**
→ Run code, debug errors, and see messages.
- 7. What is the significance of adhering to web standards in achieving cross-browser compatibility?**
→ It ensures websites work correctly across different browsers.

Git Basics

Git → A tool to track changes in code (version control).

Repository (Repo) → A folder where your project and its history are stored.

Commit → A snapshot of your code at a point in time.

Branch → A separate line of development (like working on a new feature without touching the main code).

Merge → Combining changes from one branch into another.

Clone → Copying a repo from somewhere (like GitHub) to your computer.

Pull → Get the latest changes from a remote repo.

Push → Send your changes to a remote repo.

GitHub Basics

GitHub → A platform to host Git repositories online.

Remote Repo → The version of your repo stored on GitHub.

Pull Request (PR) → A request to merge your changes into another branch (often reviewed by teammates).

Fork → Your own copy of someone else's repo (so you can experiment).

Issues → A way to track bugs, tasks, or feature requests.

Actions → Automations (like testing code whenever you push).

Other Useful Things (Git & GitHub Related)

README.md → A file that explains your project (first thing people see).

.gitignore → A file that tells Git which files/folders to ignore (like passwords or temporary files).

Open Source → Public projects where anyone can contribute.

Collaboration → GitHub makes teamwork easy: review code, suggest changes, and keep everything organized.

In short:

Git = tool for tracking code changes.

GitHub = place to share and collaborate on Git projects.

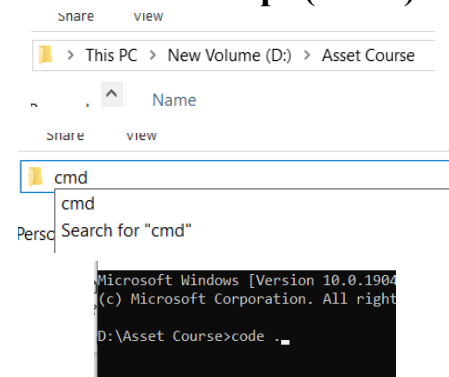
How to open a folder in Visual Studio Code using Command Prompt (CMD)?

Open Folder

Type “cmd” in the **address bar**

Type “code” [press space] . [press period/dot]”

Press “enter” to open it in **VS Code**



Frontend vs Backend vs Fullstack

Frontend (The Face)

What users **See and Interact With**

- Languages: HTML, CSS, JS
- Think: Buttons, colors, layouts, animations
- Example: When you click a button → something happens visually

In short: *Frontend = Beauty + Interaction*

Backend (The Brain)

What users **Don't See**

- Languages: Node.js, Python, PHP, Java
- Handles: Database, login system, server logic
- Example: When you log in → it checks your password

In short: *Backend = Logic + Data*

Fullstack (The Whole Machine)

Frontend + Backend combined

- You build everything
- You are the “**One-Man Army**” Dev

In short: *Fullstack = Boss Mode*

Essential VS Code Extensions

- **Live Server** → Run your website instantly
- **Prettier** → Auto-format code (makes it look clean)
- **Auto Rename Tag** → Rename one tag, it updates the pair
- **HTML CSS Support** → Helps with CSS suggestions
- **IntelliSense** → Smart auto-complete
- **Bracket Pair Colorizer** → Makes brackets colorful (lifesaver)
- **Path Intellisense** → Auto file paths
- **ESLint** → Finds errors in JS (JavaScript)
- **HTML 5**
- **CSS 3**
- **Auto Closing Tag**
- **Auto Rename Tag**
- **File and Folder Icon**
- **Code Snap**

[Reality check: Good tools don't make you a developer, but bad tools def will slow you down.]

VS Code Tips & Tricks (Speed = Power)

Keyboard Shortcuts/Hotkeys:

- **! + Tab** → Full HTML boilerplate instantly
- **Shift + Alt + F** → Format code (clean like a CEO office)
- **Ctrl + /** → Comment / Uncomment
- **Alt + Up/Down** → Move lines
- **Shift + Alt + Down** → Copy line
- **Ctrl + D** → Select same word multiple times
- **Ctrl + Space** → Trigger suggestions

[Think of shortcuts like cheat codes in this game.]

HTML Basics

(To learn more about html basics, check the “Basics of HTML” zipped folder provided in the group)

Markup Language (Like bones of a human body)

HTML = Structure of a webpage

Tags = Keywords inside < >

Example: <p>Hello</p>

Opening + Closing Tag (Like opening and closing a box)

<p>Text</p>

<p> = start

</p> = end

Self-Closing Tag (No closing needed, like a single action, done and gone)

<hr>

Head & Body

<head> → brain (info, settings)</head>

<body> → body (what users see)</body>

Meta Tags (Extra info for browser)

<meta charset="UTF-8">

UTF-8 (Without it → text breaks into nonsense)

Supports all languages (Arabic, Bangla, Chinese, etc.)

Headings & Paragraph

<h1>Big Title</h1>

<p>Text</p>

h1 = king, h6 = pawn (if you play chess then you'll get it)

Frequently Used Tags

Text & Structure

- `<h1>` to `<h6>` → Headings **`<h1>`**, **`<h2>`**, **`<h3>`**, **`<h4>`**, **`<h5>`**, **`<h6>`**
 - `<p>` → Paragraph
 - `<br>` → Line break (like pressing Enter)
 - `<hr>` → Horizontal Line
-

Text Styling

- `<b>` → Bolder text (**Like this**)
 - `<strong>` → Important (bold, serious tone, **like this**)
 - `<em>` → Emphasis (italic, emotional tone, *like this*)
 - `<i>` → Just italic (style only, *like this*)
 - `<small>` → Smaller text (somewhat similar to h6)
-

Special Meaning Tags

- `<mark>` → Highlight text (like this)
 - `<del>` → Deleted text (~~like this~~)
 - `<q>` → Short quote
 - `<blockquote>` → Long quote
-

Grouping

- `<span>` → Small inline container (Used for styling part of text)
-

Address

- `<address>` → Contact info (*somewhat similar like italic as well*)

Full Forms of Abbreviations

LAN → Local Area Network

WAN → Wide Area Network

VPN → Virtual Private Network

TCP → Transmission Control Protocol

IP → Internet Protocol

HTML → HyperText Markup Language

CSS → Cascading Style Sheets

PHP → Hypertext Preprocessor

SQL → Structured Query Language

AJAX → Asynchronous JavaScript and XML

AngularJS → Angular JavaScript (front-end framework by Google)

ReactJS → React JavaScript (UI library by Meta/Facebook)

Perl → Practical Extraction and Reporting Language

MySQL → My Structured Query Language

XAMPP → Cross-platform, Apache, MySQL, PHP, Perl

WAMP → Windows, Apache, MySQL, PHP

MAMP → Mac, Apache, MySQL, PHP

LAMP → Linux, Apache, MySQL, PHP

URL → Uniform Resource Locator

DOM → Document Object Model

PPP → Personal Protective Equipment

OSH → Occupancy Safety and Health